Synthetic PU Leather (DMF-free & DMF-based)

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Overview

Two LCA models have been developed for PU (polyurethane) synthetic leather production:

- **DMF-free (water-borne)**: Uses water as solvent — "solvent-free", lower VOCs
- **DMF-based (solvent-based)**: Uses N,N-Dimethylformamide as solvent — toxic, higher VOCs, higher environmental impact

Both models cover a combined wet and dry manufacturing process for PU leather with a thickness of 1.5–2 mm, intended for footwear applications.

LCA Design Parameters

- **Functional unit**: 1 m² PU leather (1.5–2 mm thickness)
- **System boundaries**: Cradle-to-gate
- **Geographical boundaries**: CN, GLO, RoW
- **Background database**: ecoinvent 3.12 cutoff
- **Impact assessment**: EF v3.1 (16 categories)
- **Brightway project**: `synthetic_leather`
- **Foreground databases**: `PU leather DMF-free disag 1 m2`, `PU leather DMF-based disag 1 m2`

> **Converting from 1 m² to 1 kg:** The datasets assume an areal density of **1 kg/m²** for the finished PU leather (1.5–2 mm thickness). This means 1 m² = 1 kg, and impact scores per m² can be read directly as scores per kg. This is a simplifying assumption consistent with typical PU synthetic leather in the 1.5–2 mm thickness range for footwear applications.

Manufacturing Process

The model adopts a combined wet and dry production process with the following steps:

1. **Fabric preparation**: The base fabric undergoes cleaning and surface treatment to improve adhesion and ensure uniform polymer application.
2. **Impregnation** (wet method): A polyurethane slurry, formulated with additives, is applied to the fabric using the wet method. The impregnated fabric then undergoes drying and curing to stabilize the initial PU structure.
3. **Foam layer** (wet method): A foamed polymer mixture is synthesized by incorporating thickening agents and foaming agents. This layer is uniformly applied to the material and cured at controlled temperatures to ensure proper expansion and stability.
4. **Top layer** (dry method): The final polyurethane mixture is synthesized and cast onto release paper using the dry method. After drying, it is glued or laminated onto the foam layer, finalizing the PU leather structure.

Each foreground database contains 5 activity datasets corresponding to these process steps plus a "Finished PU leather" assembly activity.

Key Differences: DMF-based vs DMF-free

The DMF-based (solvent-based) process results in:

- Use of DMF as solvent → higher toxicity impacts
- Longer drying periods (to remove DMF)
- More water consumption
- More wastewater (contaminated with DMF)

Electricity consumption: methodology and justification

Electricity is the single largest driver of the impact difference between the two leather types. The electricity values used in the inventory are **not directly reported** in the patents. Instead, they were derived through a multi-step approach:

1. **Process parameters from patents**: Both patents (CN102080332B and CN106192450A) specify process temperatures (65–100°C), stirring/drying durations, and coating steps, but do not report energy consumption directly.
2. **Scale-up methodology**: The Piccinno et al. (2016) framework was applied to convert laboratory-scale process parameters into industrial-scale energy estimates. This is a standard approach in LCA for scaling up chemical processes.
3. **Industrial machinery specs**: Energy demand was calibrated against real machinery data, including a 4.4 kW industrial curing oven (UAVOS composites oven 700–2000) and PU roller coating machine specifications.
4. **Area-based energy**: The resulting electricity values are fundamentally **area-based** (energy to coat and dry 1 m^2 of surface), which is why the m^2 functional unit is the natural choice for these datasets.

🔬 **Piccinno et al. (2016) scale-up framework — applied to PU leather**
The problem: patents describe *how* to make PU leather (temperatures, times, materials) but don't report energy consumption. We need kWh/m^2 values for the LCA inventory.

The approach in 3 steps:

1. **Extract process parameters from patents** — for each step (fabric preparation, impregnation, foam layer, top layer): process temperatures (65–153°C), stirring/drying durations, solvent types and amounts, coating thicknesses.
2. **Apply thermodynamic equations** to calculate the energy each step requires:
   - *Heating energy:* $Q = m \cdot c_p \cdot \Delta T$ (mass × specific heat capacity × temperature change)
   - *Evaporation energy:* $Q = m \cdot L_v$ (mass × latent heat of vaporization — for drying and solvent removal)
   - *Mechanical energy:* $P \times t$ (power draw of industrial machinery × operating time)
3. **Calibrate against industrial machinery specs** — instead of lab beakers and hotplates, energy demand is matched to real equipment: a 4.4 kW industrial curing oven (UAVOS 700–2000) and PU roller coating machine specifications.

Why this drives the DMF difference: DMF boils at 153°C vs water at 100°C. When you plug these into $Q = m \cdot L_v$, removing DMF from the top layer coating requires ~4× more thermal energy, plus a solvent recovery system. This is not an assumption — it follows directly from thermodynamics.

Reference: Piccinno, F. et al. (2016). From laboratory to industrial scale: a scale-up framework for chemical processes in life cycle assessment studies. Journal of Cleaner Production, 135, 1085–1097.

The resulting electricity consumption per process step (per m^2):

Process step	DMF-free (kWh/m²)	DMF-based (kWh/m²)	Difference driver
Fabric preparation	2	2	Same process
Impregnation	2.055	2.04	Similar
Foam layer	6	6.07	Similar
Top layer	**6.11**	**24**	**DMF evaporation + solvent recovery**
Total	**16.17**	**34.11**	**+111% for DMF-based**

The top layer is the only process step with a significant electricity difference. The DMF-based top layer requires ~4x more energy (24 vs 6.11 kWh/m²) because:

- **DMF has a higher boiling point** (153°C) than water (100°C), requiring higher temperatures and longer drying cycles
- **Solvent recovery** is needed to capture and recycle the evaporated DMF, adding additional energy-intensive steps
- **Extended curing time** is necessary to ensure complete DMF removal from the coating, as residual DMF is a health concern

This electricity differential in the top layer alone accounts for the majority of the ~1.9x overall impact difference between the two leather types.

Impact Assessment Results (EF 3.1, per m²)

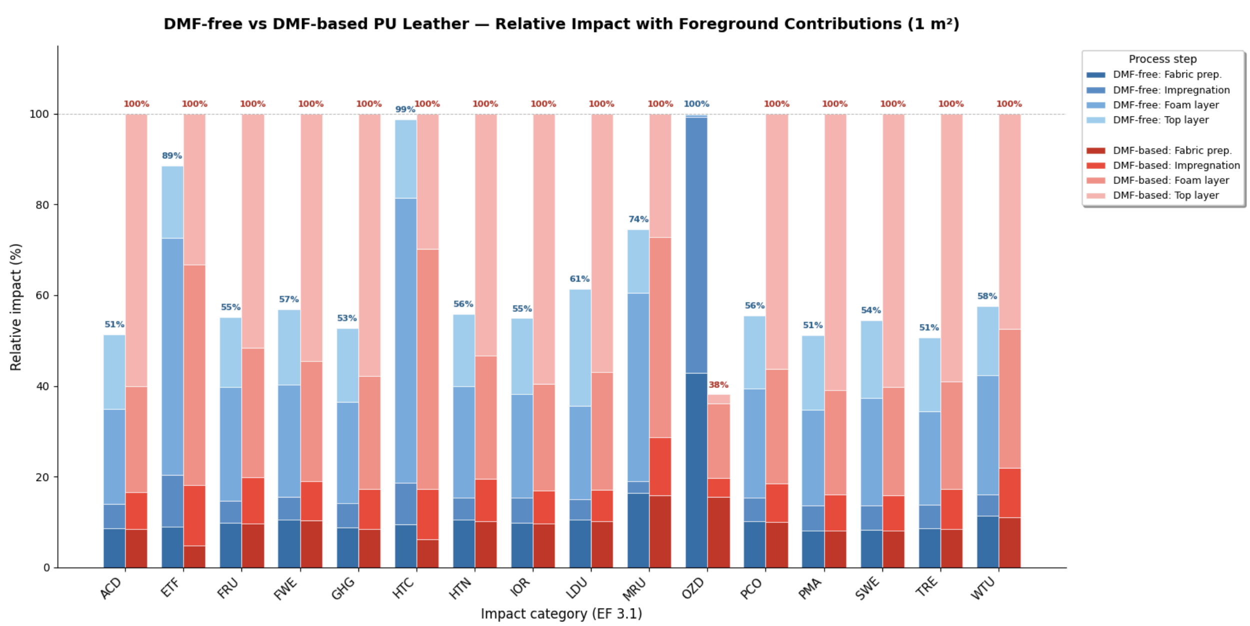
Method	Unit	DMF-free	DMF-based
ACD	mol H+ eq/m²	0.106	0.206
ETF	CTUe/m²	235.5	265.9
FRU	MJ/m²	250.6	454.2
FWE	kg P eq/m²	4.27E-03	7.50E-03
GHG	**kg CO2 eq/m²**	**19.5**	**37.0**
HTC	CTUh/m²	9.23E-09	9.35E-09
HTN	CTUh/m²	1.33E-07	2.38E-07
IOR	kBq U235 eq/m²	0.924	1.684
LDU	Pt/m²	64.2	104.7

Method	Unit	DMF-free	DMF-based
MRU	kg Sb eq/m²	6.71E-05	9.02E-05
OZD	kg CFC-11 eq/m²	2.35E-05	8.97E-06
PCO	kg NMVOC eq/m²	0.074	0.133
PMA	disease incidence/m²	1.46E-06	2.85E-06
SWE	kg N eq/m²	0.024	0.044
TRE	mol N eq/m²	0.240	0.474
WTU	m3 world eq/m²	3.50	6.08

The DMF-based process shows ~1.9x higher climate change impact (37.0 vs 19.5 kg CO2 eq/m²) and generally higher impacts across most categories, primarily due to the use of DMF solvent and longer energy-intensive drying processes.

Foreground Contribution Analysis

The chart below compares both leather types across all 16 EF 3.1 impact categories on a **relative scale** — for each category, the higher-impact leather is set to 100% and the other is expressed as a proportion. Each bar is **stacked by process step** (blue shades = DMF-free, red shades = DMF-based), revealing not only *how much* worse the DMF-based process is, but *where* in the manufacturing chain the difference originates. The dominant role of the **top layer** (lightest shade) in the DMF-based process is immediately visible across nearly all categories.



The following tables show the contribution (%) of each foreground process step to the total impact of 1 m² Finished PU leather, across all 16 EF 3.1 impact categories.

DMF-free (water-borne)

Method	Fabric preparation	Impregnation	Foam layer	Top layer
ACD	16.6	10.4	**40.9**	32.1
ETF	10.1	12.9	**59.0**	18.0
FRU	18.0	8.6	**45.5**	28.0
FWE	18.6	8.8	**43.4**	29.2
GHG	**16.6**	**10.1**	**42.5**	**30.8**
HTC	9.6	9.3	**63.6**	17.5
HTN	18.8	8.7	**44.1**	28.5
IOR	17.9	9.9	**41.6**	30.5
LDU	17.2	7.2	33.7	**41.9**
MRU	22.0	3.6	**55.7**	18.7
OZD	42.9	**56.3**	0.6	0.2
PCO	18.4	9.2	**43.3**	29.2
PMA	16.0	10.6	**41.2**	32.3
SWE	15.2	9.9	**43.4**	31.5
TRE	17.0	10.3	**40.6**	32.1
WTU	19.9	8.0	**45.8**	26.3

DMF-based (solvent-based)

Method	Fabric preparation	Impregnation	Foam layer	Top layer
ACD	8.4	8.2	23.4	**60.1**
ETF	4.8	13.3	**48.5**	33.3
FRU	9.6	10.2	28.5	**51.6**
FWE	10.3	8.7	26.6	**54.5**
GHG	**8.5**	**8.8**	**24.9**	**57.8**
HTC	6.2	11.1	**52.8**	29.9
HTN	10.3	9.2	27.1	**53.4**

Method	Fabric preparation	Impregnation	Foam layer	Top layer
IOR	9.6	7.3	23.6	**59.5**
LDU	10.2	7.0	25.9	**56.9**
MRU	15.9	12.7	**44.2**	27.2
OZD	40.9	10.9	**43.2**	5.0
PCO	10.0	8.4	25.3	**56.3**
PMA	8.1	7.9	23.1	**60.9**
SWE	8.1	7.7	23.9	**60.3**
TRE	8.5	8.7	23.8	**59.0**
WTU	11.0	11.0	30.5	**47.5**

Interpretation

DMF-free: Foam layer dominates

For the water-borne process, the **foam layer** is consistently the largest contributor across nearly all impact categories, accounting for **41–64%** of total impacts. This is driven by:

- The large input of **polyether polyols** (0.58 kg/m²) — a petroleum-derived chemical that dominates the material footprint
- Significant **electricity consumption** (6 kWh/m²) for curing and drying

The **top layer** is the second-largest contributor (18–42%), largely due to its own electricity demand (6.11 kWh/m²), kraft paper, and polyurethane adhesive. **Land use (LDU)** is the only category where the top layer exceeds the foam layer (41.9% vs 33.7%), driven by the kraft paper input.

DMF-based: Top layer dominates

For the solvent-based process, the contribution profile shifts dramatically — the **top layer** becomes the dominant contributor for most categories, accounting for **47–61%** of impacts. This is almost entirely driven by the **extremely high electricity demand of 24 kWh/m²** in this process step (vs. 6.11 kWh/m² for DMF-free), required for the extended drying cycles needed to evaporate the DMF solvent from the top coating.

The **foam layer** drops to second place (23–53%), though it still dominates for **ecotoxicity (ETF: 49%)** and **carcinogenic human toxicity (HTC: 53%)**, where the chemical composition (DMF inputs and emissions) plays a larger role than energy.

Ozone depletion: an outlier

Ozone depletion (OZD) behaves differently from all other categories in both leather types:

- **DMF-free**: Impregnation (56%) and fabric preparation (43%) dominate, while the foam and top layers contribute <1%. This is driven by upstream halon and CFC emissions in the polydimethylsiloxane and polyester supply chains.
- **DMF-based**: Foam layer (43%) and fabric preparation (41%) dominate, reflecting the shift in chemical inputs.

Key takeaway

The foreground contribution analysis highlights that **electricity consumption is the primary lever** for reducing the environmental footprint of both PU leather variants. For the DMF-based process specifically, the top layer's 24 kWh/m² electricity demand (used for extended DMF solvent evaporation and drying) is the single largest driver of the ~1.9x higher overall impact compared to the DMF-free variant. Transitioning from DMF-based to water-borne processes directly eliminates this energy penalty.

Limitations & Caveats

1. **Single patent per leather type** — The foreground inventory for each variant is based on a single Chinese patent (CN102080332B for DMF-free, CN106192450A for DMF-based). These represent specific laboratory formulations, not industry averages. Different manufacturers will use different formulations, material grades, and process conditions, which could lead to materially different results.
2. **Estimated electricity, not measured** — The electricity values (the single most influential parameter) are derived via the Piccinno et al. (2016) scale-up framework from patent process parameters and industrial machinery specs. They are engineering estimates, not measurements from an actual production line. This is the largest source of uncertainty in the model.
3. **Chinese electricity grid** — All electricity is modeled as CN medium voltage (Chinese grid mix). Because electricity dominates the impact profile, results are highly sensitive to grid carbon intensity. Using a European or renewable-heavy grid

would significantly reduce absolute impacts and narrow the gap between the two leather types.

4. **No uncertainty or sensitivity analysis** — All results are single deterministic values. No Monte Carlo simulation or sensitivity analysis has been performed on key parameters (electricity, polyol amounts, DMF recovery rates). The reported scores should be interpreted as central estimates, not precise values.
5. **Areal density assumption (1 kg/m²)** — The 1:1 conversion between m² and kg assumes an areal density of 1 kg/m² for 1.5–2 mm PU leather. Actual areal density varies with thickness, formulation, and foam density. This assumption should be validated for specific products.
6. **Cradle-to-gate system boundary** — The model covers raw material extraction through manufacturing but excludes transport to consumer, use phase, and end-of-life. This is standard for material comparison but means the results do not represent full life-cycle impacts.
7. **HTC near-identical despite DMF toxicity** — Carcinogenic human toxicity (HTC) scores are nearly identical for both types (9.23E-09 vs 9.35E-09 CTUh/m²). This is likely because EF 3.1 characterization factors may not fully capture DMF-specific toxicity (DMF is classified as a reproductive toxicant by ECHA). The HTC results should be interpreted with caution.
8. **No inbound transport** — Transport of raw materials to the manufacturing site is not included in the foreground model. In practice, this contribution is typically small relative to material production and energy use, but it is a known omission.

Data Sources

Patents

- **Patent 1 (Water-borne / DMF-free)**: Luo, X., Feng, J., Qu, J., & Li, P. (2012). *A kind of manufacture method of waterborne polyurethane synthetic leather bass*. Chinese Patent CN102080332B. Assignee: Hefei Scisky Technology Co Ltd. Google Patents
- **Patent 2 (DMF-based / Solvent-based)**: Huang, Z. (2016). *A kind of manufacturing process of wet-type PU synthetic leather*. Chinese Patent CN106192450A. Assignee: Fujian Boyi Mstar Technology Ltd. Google Patents

Scientific Literature

- **Sur, A. et al. (2018)**. Study on the manufacturing process of PU synthetic leather. *Journal of Engineering*. Wiley
- **Wang, Y. & Jin, L. (2018)**. Polyurethane-related research. *Polymers*, 10(3), 289. MDPI
- **Piccinno, F. et al. (2016)**. From laboratory to industrial scale: a scale-up framework for chemical processes in life cycle assessment studies. *Journal of Cleaner Production*, 135, 1085–1097. ScienceDirect
- **Harding, K. (2008)**. A generic approach to environmental assessment of microbial bioprocesses through life cycle assessment (LCA). ResearchGate

Industry Sources

- **Higg Index** — Synthetic leather module. Higg MSI
- **BZ Leather** — PU leather manufacturing process. bzleather.com
- **Shoemakers Academy** — How to design shoes: synthetic leather (dimensions). shoemakersacademy.com
- **Ocean Plastics (OPC)** — PU Knowledge / Wanhua. opc.com.tw

Industrial Machinery

- **Coating machine** — PU synthetic leather roller coating machine. Alibaba
- **Industrial curing oven** — Composites curing oven 700–2000. UAVOS

Inventory: PU Leather DMF-free (1 m²)

5 activities, 41 exchanges (32 linked to ecoinvent-3.12-cutoff, 9 internal)

▼ Finished PU leather (1 m²)

Assembly of all upstream process steps. Location: GLO.

Exchange	Amount	Unit	Type	Database
Finished PU leather	1	m²	production	foreground
Fabric preparation	1	m²	technosphere	foreground

Exchange	Amount	Unit	Type	Database
Impregnation	1	m^2	technosphere	foreground
Foam layer	1	m^2	technosphere	foreground
Top layer	1	m^2	technosphere	foreground

▼ Fabric preparation (1 m^2)

Base fabric cleaning and surface treatment. Location: GLO.

Exchange	Amount	Unit	Type	Location
Fabric preparation	1	m^2	production	GLO
market for polydimethylsiloxane	0.0069	kg	technosphere	RoW
market for polyvinyl chloride, suspension polymerised	0.0069	kg	technosphere	RoW
market for textile, nonwoven polyester	0.23	kg	technosphere	GLO
market for wastewater, average	-0.000207	m3	technosphere	RoW
market group for electricity, medium voltage	2	kWh	technosphere	CN
market group for tap water	0.23	kg	technosphere	GLO

▼ Impregnation (1 m^2)

PU slurry application via wet method. Location: GLO.

Exchange	Amount	Unit	Type	Location
Impregnation	1	m^2	production	GLO
market for polydimethylsiloxane	0.0142	kg	technosphere	RoW
market for polyether polyols, long chain	0.0142	kg	technosphere	RoW
market for waste polyurethane	-0.00142	kg	technosphere	RoW

Exchange	Amount	Unit	Type	Location
market for wastewater, average	-0.0004245	m3	technosphere	RoW
market group for electricity, medium voltage	2.055	kWh	technosphere	CN
market group for tap water	0.4717	kg	technosphere	GLO

▼ Foam layer (1 m²)

Foamed polymer synthesis and application. Location: GLO.

Exchange	Amount	Unit	Type	Location
Foam layer	1	m²	production	GLO
market for aniline	0.006	kg	technosphere	RoW
market for carboxymethyl cellulose, powder	0.0058	kg	technosphere	GLO
market for cellulose fibre	0.023	kg	technosphere	RoW
market for polyether polyols, long chain	0.5755	kg	technosphere	RoW
market for toluene diisocyanate	0.0173	kg	technosphere	RoW
market for waste polyurethane	-0.058	kg	technosphere	RoW
market for wastewater, average	-0.000155	m3	technosphere	RoW
market group for electricity, medium voltage	6	kWh	technosphere	CN
market group for tap water	0.1727	kg	technosphere	GLO

▼ Top layer (1 m²)

Dry method PU casting and lamination. Location: GLO.

Exchange	Amount	Unit	Type	Location
Top layer	1	m²	production	GLO
market for aniline	0.0006	kg	technosphere	RoW
market for carboxymethyl cellulose, powder	0.00058	kg	technosphere	GLO
market for cellulose fibre	0.0023	kg	technosphere	RoW
market for kraft paper	0.07	kg	technosphere	RoW
market for polyether polyols, long chain	0.05755	kg	technosphere	RoW
market for polyurethane adhesive	0.08	kg	technosphere	GLO
market for toluene diisocyanate	0.00173	kg	technosphere	RoW
market for waste polyurethane	-0.0058	kg	technosphere	RoW
market for wastewater, average	-1.55E-05	m3	technosphere	RoW
market group for electricity, medium voltage	6.11	kWh	technosphere	CN
market group for tap water	0.01727	kg	technosphere	GLO

Inventory: PU Leather DMF-based (1 m²)

5 activities, 48 exchanges (36 linked to ecoinvent-3.12-cutoff, 9 internal, 3 biosphere). Includes additional exchanges for N,N-Dimethylformamide and titanium dioxide.

▼ Finished PU leather (1 m²)

Assembly of all upstream process steps. Location: GLO.

Exchange	Amount	Unit	Type	Database
Finished PU leather	1	m²	production	foreground
Fabric preparation	1	m²	technosphere	foreground
Impregnation	1	m²	technosphere	foreground
Foam layer	1	m²	technosphere	foreground
Top layer	1	m²	technosphere	foreground

▼ Fabric preparation (1 m²)

Base fabric cleaning and surface treatment. Location: GLO. *Simpler than DMF-free (no siloxane/PVC treatment).*

Exchange	Amount	Unit	Type	Location
Fabric preparation	1	m²	production	GLO
market for textile, nonwoven polyester	0.23	kg	technosphere	GLO
market for wastewater, average	-0.000207	m3	technosphere	RoW
market group for electricity, medium voltage	2	kWh	technosphere	CN
market group for tap water	0.23	kg	technosphere	GLO

▼ Impregnation (1 m²)

PU slurry application via wet method with DMF solvent. Location: GLO.

Exchange	Amount	Unit	Type	Location
Impregnation	1	m²	production	GLO
N,N-Dimethylformamide (emission to water)	0.0304	kg	biosphere	—
market for N,N-dimethylformamide	0.3	kg	technosphere	GLO
market for polydimethylsiloxane	0.001	kg	technosphere	RoW
market for polyether	0.07	kg	technosphere	RoW

Exchange	Amount	Unit	Type	Location
polyols, long chain				
market for titanium dioxide	0.002	kg	technosphere	RoW
market for waste polyurethane	-0.0073	kg	technosphere	RoW
market for wastewater, average	-0.002736	m3	technosphere	RoW
market group for electricity, medium voltage	2.04	kWh	technosphere	CN
market group for tap water	3.04	kg	technosphere	GLO

▼ Foam layer (1 m²)

Foamed polymer synthesis with DMF solvent. Location: GLO.

Exchange	Amount	Unit	Type	Location
Foam layer	1	m²	production	GLO
N,N-Dimethylformamide (emission to water)	0.043	kg	biosphere	—
market for N,N-dimethylformamide	0.426	kg	technosphere	GLO
market for aniline	0.006	kg	technosphere	RoW
market for carboxymethyl cellulose, powder	0.067	kg	technosphere	GLO
market for cellulose fibre	0.045	kg	technosphere	RoW
market for polydimethylsiloxane	0.004	kg	technosphere	RoW
market for polyether polyols, long chain	0.403	kg	technosphere	RoW
market for titanium dioxide	0.004	kg	technosphere	RoW

Exchange	Amount	Unit	Type	Location
market for waste polyurethane	-0.04	kg	technosphere	RoW
market for wastewater, average	-0.003831	m3	technosphere	RoW
market group for electricity, medium voltage	6.07	kWh	technosphere	CN
market group for tap water	4.257	kg	technosphere	GLO

▼ Top layer (1 m²)

Dry method PU casting and lamination with DMF solvent. Location: GLO.

Exchange	Amount	Unit	Type	Location
Top layer	1	m²	production	GLO
N,N-Dimethylformamide (emission to water)	0.0043	kg	biosphere	—
market for N,N-dimethylformamide	0.0426	kg	technosphere	GLO
market for aniline	0.0006	kg	technosphere	RoW
market for carboxymethyl cellulose, powder	0.0067	kg	technosphere	GLO
market for cellulose fibre	0.0045	kg	technosphere	RoW
market for kraft paper	0.07	kg	technosphere	RoW
market for polydimethylsiloxane	0.0004	kg	technosphere	RoW
market for polyether polyols, long chain	0.0403	kg	technosphere	RoW
market for polyurethane adhesive	0.08	kg	technosphere	GLO

Exchange	Amount	Unit	Type	Location
market for titanium dioxide	0.0004	kg	technosphere	RoW
market for waste polyurethane	-0.004	kg	technosphere	RoW
market for wastewater, average	-0.000383	m3	technosphere	RoW
market group for electricity, medium voltage	24	kWh	technosphere	CN
market group for tap water	0.4257	kg	technosphere	GLO

Notebook

`import_dataset.ipynb` — Imports both foreground databases into Brightway project `synthetic_leather`, links to ecoinvent 3.12, and calculates all 16 EF 3.1 impact scores per m^2.